← Back **Technology: Application Protocols**
Graded Assignment • 30 min

Due Mar 18, 11:59 PM CET

1. What does the Application Layer expect from the Transport Layer? 1 point

- ○ The Transport Layer tells the Application Layer the geographic location of all of the routers in the Internet
- ◉ A reliable pipe that delivers data from another application across the Internet
- ○ The Transport layer dynamically transports IP addresses to all of the computers connected to a Wifi Network.
- ○ The Application sends a Domain name through the Transport Layer and gets back an IP Address
- ○ The Transport Layer accurately predicts the number of hops it will take to go across the country.

2. If you were "hacking" the Hypertext Transport Protocol using the 'telnet' command, what command would you send to the web server once you are connected to retrieve a document? 1 point

- ○ RETR
- ○ DOCU-RETR
- ○ DNLD/DOC
- ◉ GET

3. Which of the following are examples of applications in the application layer? (Choose all that apply) 1 point

- [x] Chrome web browser
- [x] Microsoft Outlook
- [ ] Ethernet port
- [ ] Fiber Optic
- [ ] Router

4. When is the Internet 100% up and working? 1 point

- ○ When more than 2% of the Internet goes down, all routers simultaneously reboot to get back to 100%
- ○ Once a year the Internet is completely rebooted and stays at 100% for about ten minutes
- ◉ It never is. It is constantly having pieces connect, fail, disconnect, reboot, etc.
- ○ At midnight GMT, every day all routers reboot and the Internet is 100% up for about ten minutes

5. Last time! What are the layers of the internet, and the order in which we structure them? 1 point

- ◉ * Application
  * Transport
  * IP
  * Link
- ○ * Application
  * Router
  * Link
  * IP
- ○ * User
  * Application
  * Link
  * Router
- ○ * Port
  * Transport
  * Application
  * Link

6. What does the browser do when you click a Hypertext Link from your current web page to another web page? 1 point

- ○ It does a relational database look up in the IMDB
- ◉ It does a Request-Response Cycle
- ○ It looks at its most recent RSS feed for the Domain Name Service and selects a Network Number
- ○ It connects to port 23 and sends the "DATA" command

7. What does port 23 do? 1 point

- ○ IMAP
- ○ POP
- ○ YouTube
- ◉ Telnet (Login)

8. What does port 80 do? 1 point

- ○ Telnet (login)
- ○ SMTP
- ○ HTTPS
- ◉ HTTP

9. RFCs are: (Choose all that apply) 1 point

- ◉ The standards defining protocols on the Internet
- ○ Curated by Paul Kunz at the Stanford Linear Accelerator (SLAC)
- ○ Papers written by Physicists at CERN about Radio Frequency Colliders
- ○ Issued by the United Nations (UN)

10. Which of the following is most like a TCP/IP port number: 1 point

- ○ Telephone area code
- ○ Zip Code
- ○ Highway number
- ◉ Telephone extension

11. What is a protocol? 1 point

- ◉ A set of rules that govern how different components of the Internet interact with each other
- ○ A program used to scan for vulnerable ports on a network-connected computer
- ○ A technique routers use to predict the traffic distribution over a long time period
- ○ A technique for obscuring security algorithms